


Karunya INSTITUTE OF TECHNOLOGY AND SCIENCES

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Karunya Nagar, Coimbatore - 641 114, Tamil Nadu, India.

School of Computer Science and Technology
Division of Computer Science and Engineering
Part – A

MICRO-PROJECT PROPOSAL

I. Details of the Students:

S.No	Name of the Student	Register Number
1	Ashwin. P	(URK25CS1195)
2	Lesliy Robinson. A	(URK25CS1102)
3	Roshan. J	(URK25CS1010)

II. Details of the Course related to the Micro-Project:

Course Title:	PYTHON PROGRAMMING
Course Code:	25CS209
Course Faculty:	Dr. R VENKATESAN
Academic Year:	2025 - 2029
Semester:	II
Year & Batch:	1 st & 1

III. Macro-Level and Meso-Level Sub-Tasks (COs) related to this Micro-Project:

Macro-Level Competency:	Evaluate the functionality of application in various situations.	
Course Outcomes:	CO1	<i>Use</i> core Python concepts to build structured program components.
	CO2	<i>Use</i> Python data structures to organize data.
	CO3	<i>Develop</i> modular and reusable applications using Object Oriented Programming.
	CO4	<i>Develop</i> File handling, GUI and database-connected applications
	CO5	<i>Execute</i> image processing tasks using python libraries

Fluid Flow in a Jet Nozzle using Vector Calculus

1.0 **Rationale** *(Students should elaborate on the importance/significance of doing this project, in about 200 to 400 words.)*

The micro-project titled "Fluid Flow in a Jet Nozzle using Vector Calculus" focuses on understanding the fundamental behavior of fluid flow in jet nozzles, which are widely used in aerospace, aerodynamics, propulsion systems, and fluid machinery industries to generate thrust. Jet nozzles are essential devices used to accelerate fluid pressure to high velocities and convert pressure energy into kinetic energy, enabling thrust generation in aircraft engines and rockets.

This project is significant because it bridges theoretical mathematical concepts with real-world engineering applications. Vector calculus provides powerful tools such as velocity vectors (speed and direction control), gradient (pressure variance), divergence (spread or shrink flow), and continuity equations (conservation of mass), which are essential to describe and analyze fluid motion. By simulating these parameters, students develop analytical thinking and gain early exposure to simulation techniques similar to Computational Fluid Dynamics (CFD), commonly used in the industry for design and optimization.

Academically, the project supports the attainment of course outcomes CO2 (data structures), CO3 (Object-Oriented Programming), and CO5 (image processing/visualization). Overall, this project builds a strong foundation in fluid dynamics and aerospace engineering concepts while applying modern computational tools like Python.

2.0 **Literature Review** *(Students should provide the relevant knowledge/information and existing status, from journals/technical magazines, brochures, websites and other sources referred/required to complete the selected micro-project in about 200 to 500 words.)*

Jet nozzle fluid dynamics has been extensively studied using computational methods, focusing on vector-based flow analysis for thrust optimization. Recent research by Atmaca (2021) utilized CFD simulations to analyze exit velocities in rectangular nozzles, revealing how geometry impacts flow characteristics like maximum velocity—insights directly applicable to vector field modelling.

In terms of Python-based simulations, Stam's "Stable Fluids" (2003) remains a foundational work, allowing for real-time 2D CFD using Navier-Stokes solvers. These implementations often align with Course Outcome CO2 by using lists for grids and dictionaries for fields, and CO5 by using OpenCV for quiver plots to visualize velocity vectors. Further advancements by researchers like Ravi Ram demonstrate the use of Python for generating structured meshes and simulating rocket nozzles (e.g., PSLV), emphasizing the use of Object-Oriented Programming (OOP) for nozzle geometries.

Current industry trends, such as METU's CFD optimization (2024), highlight the necessity of vector calculus—specifically divergence-free fields—in modeling subsonic jets. These studies inform our proposed simulation, where an OOP Nozzle class will process vector data structures and visualize results through image enhancement techniques. This project aims to fill the gap in accessible Python prototypes for students, providing real-time visuals for educational thrust prediction.

3.0 **Proposed Methodology** *(Students should provide a detailed Flowchart/Procedure in brief (about 200 to 500) that will be followed to complete the micro-project.)*

The methodology for this project is structured to systematically model fluid behavior through a jet nozzle by integrating theoretical fluid mechanics with computational analysis.

Procedure:

1. **Geometric Configuration:** Define the jet nozzle design (convergent or convergent-divergent) and establish assumptions such as steady, incompressible flow to simplify mathematical modeling.
2. **Mathematical Formulation:** Formulate the governing equations—Continuity, Momentum (Navier-Stokes), and Energy—using vector calculus. This includes using the **Area of a Circle** ($A = \pi r^2$) for cross-sections and applying **Bernoulli's Theorem** and the **Equation of Continuity** ($A_1 v_1 = A_2 v_2$) to determine pressure and velocity shifts.
3. **Vector Calculus Application:** Apply vector operators such as **gradient** (for pressure distribution), **divergence** (for mass conservation), and **curl** to describe the velocity fields within the nozzle.

4. **Computational Implementation (Python & MongoDB):**
 - Develop a Python script using **NumPy** for numerical calculations and **Matplotlib** for visualization.
 - Use **OOP principles** (CO3) to create classes for the nozzle and fluid particles.
 - Implement **MongoDB connectivity** to store and retrieve simulation parameters and historical result data for performance analysis.
5. **Validation and Reporting:** Compare the computational results (velocity amplification, pressure drop) with standard theoretical characteristics to validate the model. Finally, document the findings in a structured report.

4.0 Tentative Action Plan *(Sequence of activities and time (Gantt Chart) required to complete the project.)*

S. No.	Details of the planned activities during the entire semester	Planned Start date	Planned Finish date	Name of the Responsible Team Member(s)
1	Topic selection, problem definition, and scope finalization	Second half of January	Second half of January	All Members
2	Literature review and collection of research papers/technical references	Second half of January	Second half of January	Roshan
3	Study of vector calculus concepts (Gradient, Divergence, Continuity)	First half of February	First half of February	Robin
4	Nozzle geometry modeling and defining mathematical assumptions	First half of February	First half of February	Ashwin
5	Mathematical formulation of flow equations (Bernoulli & Continuity)	First half of February	First half of February	Robin & Ashwin
6	Python backend development and environment setup (NumPy, Matplotlib)	First half of February	First half of February	Roshan & Ashwin
7	Implementation of MongoDB connectivity for simulation data storage	Second half of February	Second half of February	Roshan & Robin
8	Visualization of velocity and pressure fields using Python libraries	Second half of February	Second half of February	All Members
9	Result analysis, validation against theoretical data, and troubleshooting	First half of March	First half of March	All Members
10	Final report preparation, documentation, and project submission	First half of March	First half of March	All Members

5.0 Tentative Resources (if any) Required *(major resources such as raw materials, some machining facilities, software etc.)*

S. No.	Name of Resource/Material	Broad Specifications	Qty.	Cost (INR)
1	Laptop / Computer	Minimum 8 GB RAM, i5 Processor	1	Existing
2	Python Software	Python 3.x, NumPy, Matplotlib	1	Free
3	Internet Facility	Research & reference access	1	Existing

